

Phenotypic Expansion in Homo sapiens: Structural Coherence Under Improved Boundary Conditions

Abstract

Over the last century, Homo sapiens has exhibited rapid and widespread changes in average height, limb proportions, skeletal robustness, facial symmetry, and general physiological harmony. These changes occur on a timescale incompatible with classical Darwinian evolution driven by mutation and natural selection. This paper proposes that the observed transformations are better explained as a phase shift in phenotypic expression resulting from altered boundary conditions, within the framework of the Theorem of Axiomatic Necessity (TNA). Under this model, modern humans are not evolving genetically, but expressing latent structural potential previously suppressed by energetic and environmental constraints.

1. Evolution vs. Phenotypic Liberation

Classical evolutionary mechanisms operate effectively on genetic variation across many generations. However, genomic studies indicate that the human genome has remained largely stable over the last several thousand years.

Yet, during the same period, humans have become:

- significantly taller,
- more gracile yet mechanically efficient,
- more symmetrical in craniofacial structure,
- less developmentally stressed.

This discrepancy suggests that the dominant driver is not genetic innovation, but **phenotypic expansion under relaxed constraints**.

Within TNA, this is interpreted not as adaptation through selection, but as **structural coherence emerging once limiting boundary conditions are removed**.

2. Historical Constraint Regimes

For most of human history, Homo sapiens operated under severe energetic and environmental restrictions:

- chronic caloric scarcity,
- high pathogen load,
- micronutrient deficiencies,
- physical trauma and early mortality.

Under these conditions, the human body converged on compromise solutions prioritizing survival efficiency over structural optimization. Shorter stature, reduced limb length, and robust but coarse morphology were not optimal designs, but **necessitated configurations under scarcity**.

These forms represent **low-coherence equilibria**, not genetic ceilings.

3. Boundary Condition Shift in the Modern Era

The last 150 years introduced an unprecedented change in boundary conditions:

- sustained caloric abundance,
- medical reduction of developmental stress,
- improved sanitation and disease control,
- reduced childhood mortality.

This shift removed long-standing structural constraints. Under TNA, when boundary conditions change discontinuously, biological systems reorganize toward higher-coherence configurations without requiring genetic change.

The result is a **rapid phenotypic expansion** toward mechanically and energetically optimal human form.

4. Height and Structural Scaling

The global increase in average height illustrates this mechanism clearly.

Height expansion is not free: it requires proportional scaling of skeletal support, joint geometry, and balance architecture. Increased foot size and limb length are not anomalies but **necessary structural recalibrations** to maintain stability and center-of-mass control in taller bodies.

Such changes are coherent responses dictated by physical constraints, not selective preferences.

5. Aesthetic Perception as Coherence Detection

Human perception of attractiveness correlates strongly with indicators of low developmental stress:

- symmetry,
- clear skin,
- aligned dentition,
- proportional body geometry.

Within the TNA framework, these features are not aesthetic goals but **signals of internal coherence**. As environmental noise decreases, organisms express forms closer to their optimal structural solution.

What is perceived culturally as increased “beauty” is, biologically, the visual manifestation of reduced constraint and higher system coherence.

6. Not a New Species, but a New Operating Regime

The current transformation does not constitute speciation. *Homo sapiens* remains genetically *Homo sapiens*.

However, it represents a shift toward a **high-coherence phenotypic regime**, comparable to moving from a constrained solution to a near-optimal one within the same design space.

In TNA terms, modern humans are operating closer to their maximal feasible coherence under present boundary conditions.

7. Implications

This model implies:

- Rapid human morphological change does not require rapid evolution.
 - Darwinian selection explains local variation but not global phenotypic shifts under systemic constraint removal.
 - Biological form is partially pre-specified by structural necessity, revealed when environmental suppression is lifted.
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8. Conclusion

The modern human phenotype is not evidence of accelerated evolution, but of **structural release**.

As boundary conditions improved, Homo sapiens expanded into a configuration that had always been physically permissible but historically inaccessible. The current era represents the highest phenotypic coherence achieved by our species to date—not because we evolved into it, but because we were finally allowed to express it.

Zenodo-safe closing statement

■ *When constraints vanish, structure speaks.*